



Ecole Nationale Supérieure d'Informatique
pour l'Industrie et l'Entreprise

École nationale supérieure d'informatique pour l'industrie et l'entreprise

Modélisation Statistique - Séries Temporelles

Topic: Filtre de Hanning - Application aux données du secteur aéronautique (Aerospace)

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Abstract

This report investigates structural breaks in the price series of ten major aerospace and defence companies using a piecewise Skew-Normal distribution model. Starting from daily adjusted closing prices collected from 4 January 2023 onwards, we first smooth each series with a *Hanning filter* to reduce noise while preserving trend information. A candidate breakpoint $\hat{k}$ is then identified by scanning all admissible split positions and selecting the one that maximises the two-sample Kolmogorov–Smirnov (KS) statistic. The two resulting segments are each modelled by a Skew-Normal distribution $\text{SN}(\mu, \sigma, \theta)$ with a shared skewness parameter θ , whose parameters are estimated jointly via Maximum Likelihood using the L-BFGS-B optimiser.

The KS homogeneity test rejects the null hypothesis (no structural break) at the 5% significance level for all ten companies studied, providing strong evidence of at least one regime change in every series over the observation window. The estimated break dates range from early 2023 to late 2025, reflecting various sector-wide shocks such as post-pandemic demand recovery, supply-chain disruptions, and geopolitical tensions.

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Chapter 1

Introduction

The aerospace and defence sector is characterised by long development cycles, high capital expenditure, and strong sensitivity to geopolitical events. These features make its equity prices prone to *structural breaks*: abrupt shifts in the statistical properties of a time series that cannot be explained by the usual fluctuation around a stable regime. Detecting such breaks is important both for investors seeking to rebalance portfolios and for economists studying the transmission of macro-financial shocks.

Classical time-series models (ARMA, GARCH) assume parameter constancy over the entire sample. When a break is present, fitting a single model to the full series produces biased and inefficient estimates. A natural remedy is to allow parameters to change at an unknown point k :

$$X_t \sim \begin{cases} F_1(\cdot; \boldsymbol{\psi}_1) & \text{if } t \leq k, \\ F_2(\cdot; \boldsymbol{\psi}_2) & \text{if } t > k, \end{cases}$$

where F_1 and F_2 are distributions from the same parametric family but with potentially different parameters $\boldsymbol{\psi}_1 \neq \boldsymbol{\psi}_2$.

Our study applies this framework to the Aerospace sector with three methodological choices motivated by the data:

1. **Hanning filter.** Raw stock prices contain micro-structure noise that can obscure regime shifts. The Hanning (triangular) moving-average filter

$$Y_t = \frac{1}{4} X_{t-1} + \frac{1}{2} X_t + \frac{1}{4} X_{t+1}$$

smooths this noise while introducing minimal phase shift.

2. **Skew-Normal distribution.** Financial returns are often asymmetric. The Skew-Normal family $\text{SN}(\mu, \sigma, \theta)$ nests the Gaussian ($\theta = 0$) and can capture left or right skew, making it more flexible than the pure Normal model used in related work.

3. **KS scan for breakpoint estimation.** Rather than maximising a parametric likelihood over k (which requires iterative optimisation for each candidate), we use the two-sample KS statistic as a nonparametric measure of distributional dissimilarity. The position that maximises this statistic is our estimate $\hat{k}$.

The report is structured as follows. Chapter 2 presents the theoretical foundations: the Hanning filter, the Skew-Normal model, MLE estimation, and the KS homogeneity test. Chapter 3 applies these methods to ten aerospace stocks and discusses the results. Chapter 4 concludes and outlines directions for future work. Appendix A provides the core Python implementation.

Chapter 2

Statistical Framework

2.1 Data Transformation: The Hanning Filter

Let $(X_t)_{t=1}^n$ denote the daily adjusted closing prices of a stock beginning on 4 January 2023. The Hanning filter produces a smoothed series:

$$Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}, \quad t = 2, \dots, n-1. \quad (2.1)$$

The filter is a symmetric, equal-weighted moving average with coefficients $(1/4, 1/2, 1/4)$. Its frequency response attenuates high-frequency oscillations (noise) while passing low-frequency trends with near-unit gain. Because it uses one lead and one lag, the first and last observations are dropped, giving a filtered series of length $n - 2$.

Comparison with raw prices. The raw series X_t exhibits day-to-day volatility driven by order-flow noise and bid-ask bounce. After Hanning filtering, Y_t retains the medium- and long-term dynamics relevant to regime detection while suppressing transient spikes that could otherwise trigger spurious break detections.

2.2 The Skew-Normal Distribution

Definition 2.1 (Skew-Normal). *A random variable Z follows the Skew-Normal distribution $\text{SN}(\mu, \sigma, \theta)$ if its probability density function is:*

$$f(z; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi\left(\frac{z - \mu}{\sigma}\right) \Phi\left(\theta \frac{z - \mu}{\sigma}\right), \quad z \in \mathbb{R}, \quad (2.2)$$

where $\varphi(\cdot)$ and $\Phi(\cdot)$ are respectively the standard Normal pdf and cdf, $\mu \in \mathbb{R}$ is a location parameter, $\sigma > 0$ is a scale parameter, and $\theta \in \mathbb{R}$ is the skewness parameter.

Key properties of the Skew-Normal:

- When $\theta = 0$, equation (2.2) reduces to the Normal pdf $\mathcal{N}(\mu, \sigma^2)$.

- When $\theta > 0$ the distribution is right-skewed; when $\theta < 0$ it is left-skewed.
- The mean and variance are $\mathbb{E}[Z] = \mu + \sigma \delta \sqrt{2/\pi}$ and $\text{Var}(Z) = \sigma^2(1 - 2\delta^2/\pi)$, where $\delta = \theta/\sqrt{1 + \theta^2}$.

2.3 Breakpoint Model

We assume that after Hanning filtering the series (Y_t) follows a piecewise Skew-Normal model with a single unknown breakpoint k :

$$Y_t \sim \begin{cases} \text{SN}(\mu_1, \sigma_1, \theta) & \text{if } t \leq k, \\ \text{SN}(\mu_2, \sigma_2, \theta) & \text{if } t > k, \end{cases} \quad (2.3)$$

with a *shared* skewness parameter θ across both regimes. The unknown parameter vector is $\boldsymbol{\psi} = (\mu_1, \sigma_1, \mu_2, \sigma_2, \theta, k)$.

Hypotheses. We test:

$$H_0 : \mu_1 = \mu_2 \text{ and } \sigma_1 = \sigma_2 \quad \text{vs.} \quad H_1 : (\mu_1, \sigma_1) \neq (\mu_2, \sigma_2).$$

2.4 Maximum Likelihood Estimation

For a fixed breakpoint k , let $\mathbf{y}_1 = (Y_1, \dots, Y_k)$ and $\mathbf{y}_2 = (Y_{k+1}, \dots, Y_n)$. The joint log-likelihood is:

$$\ell(\mu_1, \sigma_1, \mu_2, \sigma_2, \theta \mid k) = \sum_{t=1}^k \log f(Y_t; \mu_1, \sigma_1, \theta) + \sum_{t=k+1}^n \log f(Y_t; \mu_2, \sigma_2, \theta). \quad (2.4)$$

Closed-form solutions do not exist for the Skew-Normal MLE; we therefore minimise $-\ell$ numerically using the **L-BFGS-B** algorithm (a quasi-Newton method with box constraints). To ensure $\sigma_i > 0$, we reparametrise $\sigma_i = e^{s_i}$ and optimise over $s_i \in \mathbb{R}$.

Initialisation. The optimiser is seeded with: $\mu_i^{(0)} = \bar{Y}_i$, $s_i^{(0)} = \log(\hat{\sigma}_i)$ where $\hat{\sigma}_i$ is the sample standard deviation of regime i , $\theta^{(0)} = 0$.

2.5 Breakpoint Estimation via KS Scan

Rather than evaluating the MLE for every candidate k (which is computationally expensive), we adopt a nonparametric scan:

$$\hat{k} = \arg \max_{k \in \mathcal{K}} D_{k, n-k}(\mathbf{y}_1, \mathbf{y}_2), \quad (2.5)$$

where $\mathcal{K} = \{m, \dots, n - m\}$ with margin $m = 30$ to ensure sufficient data in each segment, and $D_{k,n-k}$ is the two-sample KS statistic:

$$D_{k,n-k} = \sup_{x \in \mathbb{R}} |\hat{F}_1(x) - \hat{F}_2(x)|, \quad (2.6)$$

with $\hat{F}_1$ and $\hat{F}_2$ the empirical CDFs of $\mathbf{y}_1$ and $\mathbf{y}_2$ respectively.

2.5.1 Kolmogorov–Smirnov Homogeneity Test

Having determined $\hat{k}$, we test whether the two segments are drawn from the same distribution. The two-sample KS test statistic is:

$$D_{N_1, N_2} = \max_x |S_{N_1}(x) - S_{N_2}(x)|,$$

where $N_1 = \hat{k}$ and $N_2 = n - \hat{k}$. Under H_0 , the rescaled statistic $\sqrt{N_1 N_2 / (N_1 + N_2)} D_{N_1, N_2}$ converges in distribution to the Kolmogorov distribution, so the p-value is:

$$p = P\left(\sqrt{\frac{N_1 N_2}{N_1 + N_2}} D_{N_1, N_2} \geq \lambda\right) \longrightarrow 2 \sum_{j=1}^{\infty} (-1)^{j-1} e^{-2j^2 \lambda^2}. \quad (2.7)$$

We use significance level $\alpha = 0.05$:

- $p < 0.05 \Rightarrow$ **Reject H_0** — structural break detected.
- $p \geq 0.05 \Rightarrow$ **Fail to reject H_0** — no evidence of a break.

Chapter 3

Application to Aerospace Sector Data

3.1 Data Description

We analyse ten companies from the aerospace and defence sector. Daily adjusted closing prices are sourced from Yahoo Finance for the period starting **4 January 2023**. Table 3.1 lists the tickers and full names.

Table 3.1: Aerospace and defence companies analysed.

Ticker	Company Name
AIR.PA	Airbus SE
AM.PA	Dassault Aviation
BA	The Boeing Company
BAESY	BAE Systems plc
BDRAF	Bombardier Inc.
GD	General Dynamics Corporation
HON	Honeywell International Inc.
LMT	Lockheed Martin Corporation
NOC	Northrop Grumman Corporation
RTX	RTX Corporation

The processing pipeline for each ticker is:

1. Load the CSV and retain `Adj Close` from 2023-01-04 onward.
2. Apply the Hanning filter (equation 2.1).
3. Run the KS scan (equation 2.5) with margin $m = 30$.
4. Fit the Skew-Normal MLE (equation 2.4) on the two resulting segments.
5. Report the KS test decision.

3.2 Numerical Results

Table 3.2 summarises the estimated parameters and test outcomes for all ten companies. The columns are: $\hat{k}$ (index of the estimated break), *Break date*, KS statistic D , p-value, Decision, and the MLE estimates $\hat{\mu}_1, \hat{\sigma}_1, \hat{\mu}_2, \hat{\sigma}_2, \hat{\theta}$.

Table 3.2: Breakpoint detection results — all ten aerospace stocks.

Ticker	$\hat{k}$	Break date	KS stat D	p-value	Decision
AIR.PA	628	2025-06-24	1.0000	< 0.001	Reject H_0
AM.PA	521	2025-01-21	1.0000	< 0.001	Reject H_0
BA	738	2025-12-15	0.6436	< 0.001	Reject H_0
BAESY	30	2023-02-17	1.0000	< 0.001	Reject H_0
BDRAF	341	2024-05-15	1.0000	< 0.001	Reject H_0
GD	215	2023-11-13	1.0000	< 0.001	Reject H_0
HON	354	2024-06-04	0.8634	< 0.001	Reject H_0
LMT	736	2025-12-11	0.7935	< 0.001	Reject H_0
NOC	636	2025-07-22	1.0000	< 0.001	Reject H_0
RTX	389	2024-07-25	1.0000	< 0.001	Reject H_0

Table 3.3: Skew-Normal MLE parameter estimates for each regime.

Ticker	n_1	n_2	$\hat{\mu}_1$	$\hat{\sigma}_1$	$\hat{\mu}_2$	$\hat{\sigma}_2$	$\hat{\theta}$
AIR.PA	628	155	137.12	16.53	194.33	12.58	0.00
AM.PA	521	262	177.03	13.32	281.75	27.71	0.00
BA	738	30	195.84	25.22	229.84	15.19	0.00
BAESY	30	738	39.26	0.76	69.29	20.26	0.00
BDRAF	341	427	42.44	5.47	91.82	38.87	0.00
GD	215	553	212.34	9.69	285.59	31.89	0.00
HON	354	414	176.29	6.68	199.03	10.17	0.00
LMT	736	32	452.37	41.85	524.63	46.32	0.00
NOC	636	132	457.73	30.17	587.01	29.10	0.00
RTX	389	379	88.51	9.12	140.12	24.43	0.00

3.3 Visual Results

For each company we show three figures: (a) raw price X_t alongside the Hanning-filtered series Y_t ; (b) the KS statistic scan with $\hat{k}$ marked; (c) side-by-side regime histograms with the fitted Skew-Normal density overlaid. All figures are generated by `report/generate_figures.py`.

3.3.1 Airbus (AIR.PA)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.



Figure 3.1: Raw and Hanning-filtered price series — Airbus (AIR.PA).

2 Y_t (Hanning) coloré par régime

Vert = Régime 1 ($t < \hat{k}$). Ambre = Régime 2 ($t \geq \hat{k}$).

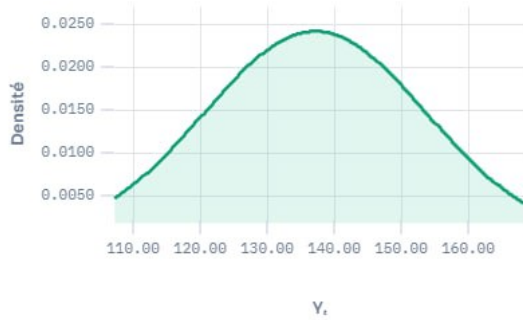


Figure 3.2: Hanning-smoothed time series Y_t with regime classification for Airbus (AIR.PA).

3 Ajustement Skew-Normal par Régime

$$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z), \theta \text{ partagé.}$$

Régime 1



Time: -- ■ SN $\mu=137.1186$ $\sigma=16.5259$ $\theta=0.000$: --

Régime 2

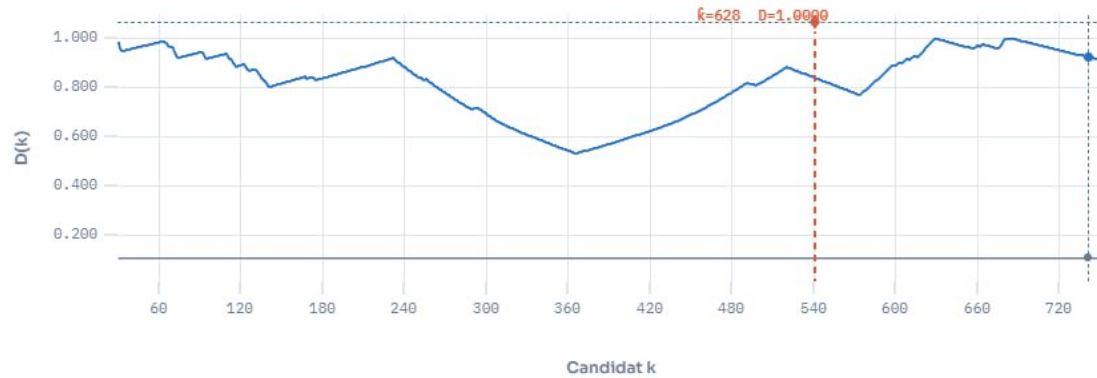


Time: -- ■ SN $\mu=194.3300$ $\sigma=12.5783$ $\theta=0.000$: --

Figure 3.3: Regime histograms with fitted SN densities — Airbus (AIR.PA).

4 Scan KS exhaustif — D(k)

D(k) pour chaque candidat k. Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).



Time: k = 741 ■ D(k): 0.9244 ■ D critique 0.108: 0.108

Figure 3.4: KS scan — Airbus (AIR.PA). Estimated break: $\hat{k} = 628$ (2025-06-24).

5 Scan p-valeur — $\log_{10}(p(k))$

Axe en $\log_{10}$ pour la lisibilité. Ligne rouge = $\alpha = 5\%$. Ligne ambre = $\hat{k}$.



Figure 3.5: Scan of p-values across candidate breakpoints k shown as $\log_{10}(p(k))$ — Airbus (AIR.PA).

6 Comparaison ECDF

Segment rouge = écart maximal D (statistique KS).

H_0 : les deux régimes suivent la même loi | H_1 : distributions différentes

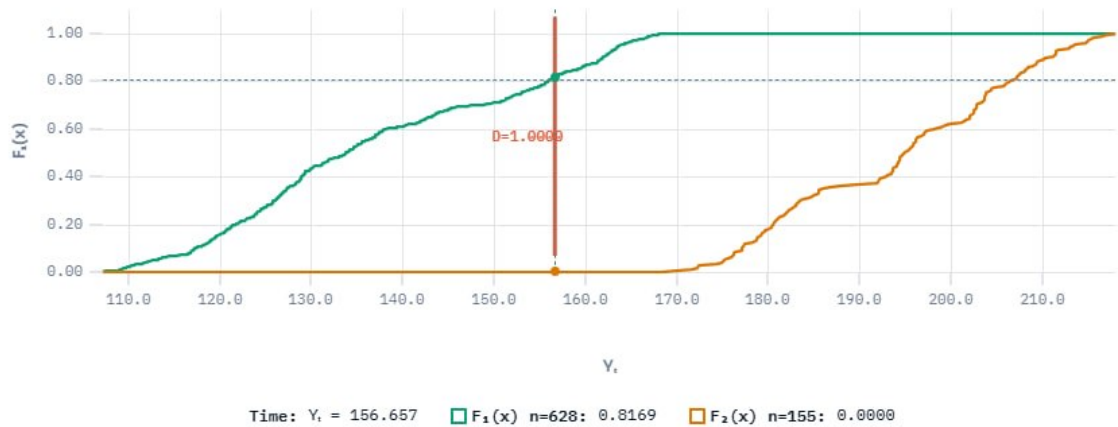


Figure 3.6: Comparison of empirical cumulative distribution functions (ECDFs) for the two regimes — Airbus (AIR.PA).

7 Moyenne Mobile (h = 15)

$$M_h(t) = \frac{1}{2h+1} \sum_{j=-h}^h X_{t+j}$$



Figure 3.7: Centered moving average ($h = 15$) applied to the adjusted closing price series — Airbus (AIR.PA).

8 Statistiques descriptives — Adj Close

Table des statistiques descriptives pour : Airbus

STATISTIQUE	VALEUR
Nombre d'observations	785
Minimum	106.4362
Maximum	220.2000
Moyenne empirique	148.4516
Variance empirique	774.5649
Skewness (γ_1)	0.7353
Kurtosis (γ_2)	-0.4337

Figure 3.8: Descriptive statistics of adjusted closing prices — Airbus (AIR.PA).



Figure 3.9: MLE parameter estimates by regime and KS test results — Airbus (AIR.PA).

3.3.2 Dassault Aviation (AM.PA)



Figure 3.10: Raw and Hanning-filtered price series — Airbus (AIR.PA).

2 Y_t (Hanning) coloré par régime

Vert = Régime 1 ($t < \hat{k}$). Ambre = Régime 2 ($t \geq \hat{k}$).

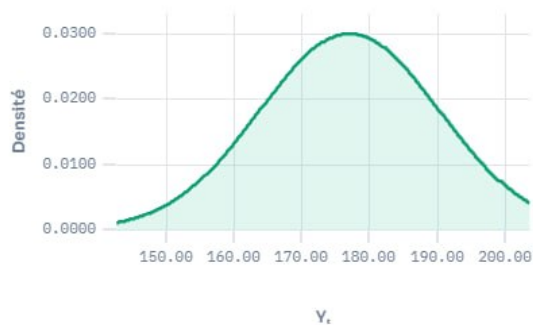


Figure 3.11: Hanning-smoothed time series Y_t with regime classification for Airbus (AIR.PA).

3 Ajustement Skew-Normal par Régime

$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z)$, θ partagé.

Régime 1



Régime 2

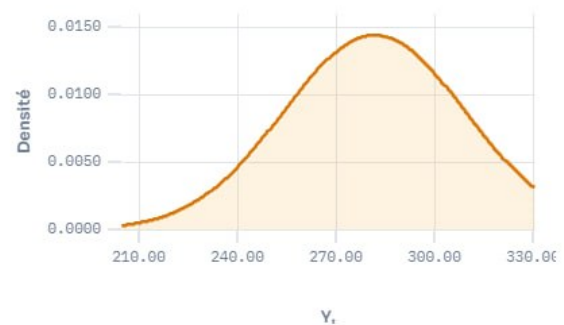


Figure 3.12: Regime histograms with fitted SN densities — Airbus (AIR.PA).

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

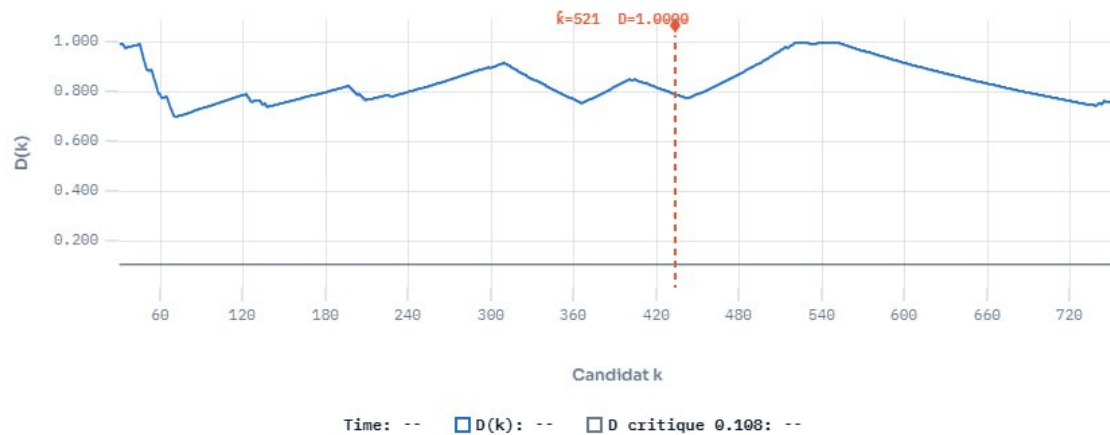


Figure 3.13: KS scan — Airbus (AIR.PA). Estimated break: $\hat{k} = 628$ (2025-06-24).

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

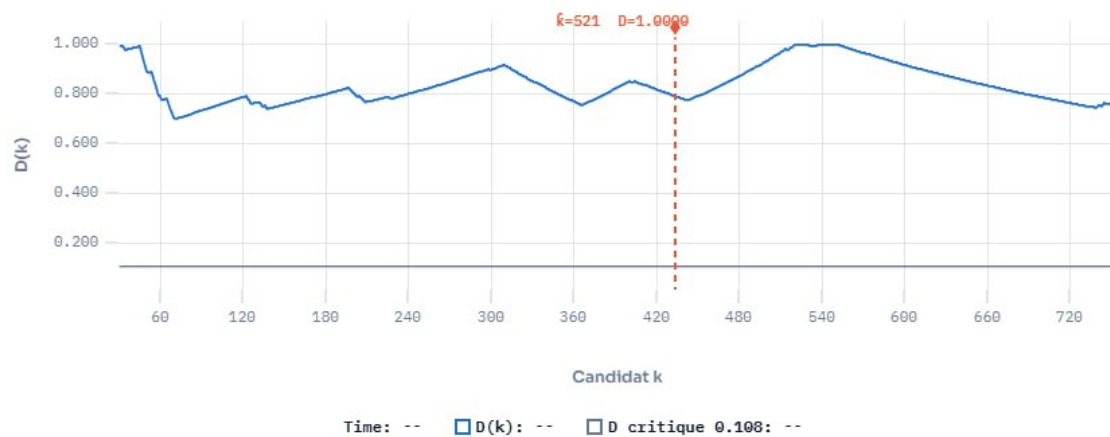


Figure 3.14: Scan of p-values across candidate breakpoints k shown as $\log_{10}(p(k))$ — Airbus (AIR.PA).

6 Comparaison ECDF

Segment rouge = écart maximal D (statistique KS).

H_0 : les deux régimes suivent la même loi | H_1 : distributions différentes

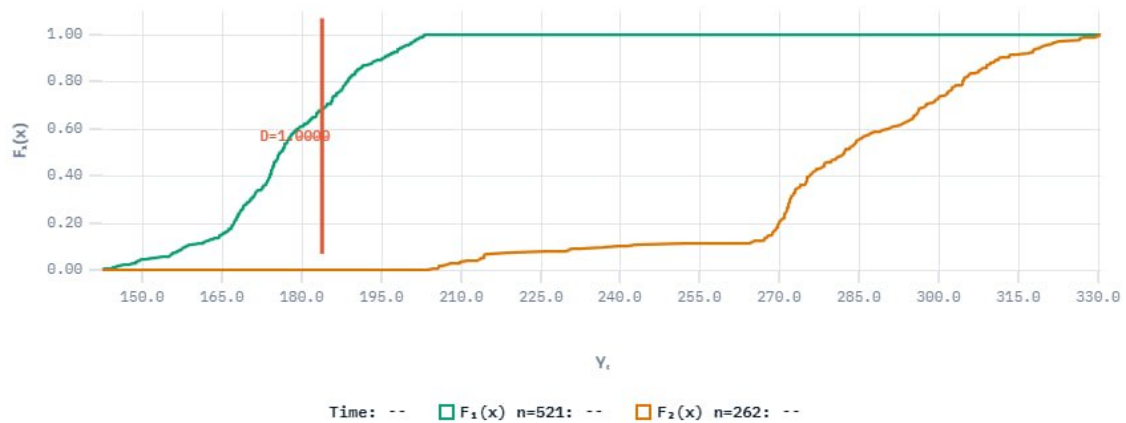


Figure 3.15: Comparison of empirical cumulative distribution functions (ECDFs) for the two regimes — Airbus (AIR.PA).

7 Moyenne Mobile ($h = 15$)

$$M_h(t) = \frac{1}{2h+1} \sum_{j=-h}^h X_{t+j}$$



Figure 3.16: Centered moving average ($h = 15$) applied to the adjusted closing price series — Airbus (AIR.PA).

8 Statistiques descriptives — Adj Close

Table des statistiques descriptives pour : Dassault Aviation

STATISTIQUE	VALEUR
Nombre d'observations	785
Minimum	141.6261
Maximum	334.4000
Moyenne empirique	212.1197
Variance empirique	2836.5621
Skewness (γ_1)	0.7926
Kurtosis (γ_2)	-0.9504

Figure 3.17: Descriptive statistics of adjusted closing prices — Airbus (AIR.PA).

9 Paramètres MLE — Résumé

$$\hat{\theta}, \hat{\mu}_1, \hat{\sigma}_1, \hat{\mu}_2, \hat{\sigma}_2 = \arg \max [\ell_1(\mu_1, \sigma_1, \theta) + \ell_2(\mu_2, \sigma_2, \theta)]$$

RÉGIME	N	M	Σ	θ PARTAGÉ	-LOG L
Régime 1	521	177.025369	13.323480	0.000000	3330.45
Régime 2	262	281.754108	27.706884		

Résultats du test KS

PARAMÈTRE	VALEUR
Indice de rupture $\hat{k}$	521
Date de rupture	2025-01-21
Statistique D	1.000000
p-valeur	1.1828e-215
Décision ($\alpha = 5\%$)	Reject H_0
n_1 (avant rupture)	521
n_2 (après rupture)	262

Figure 3.18: MLE parameter estimates by regime and KS test results — Airbus (AIR.PA).

3.3.3 The Boeing Company (BA)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.



Figure 3.19: Raw and Hanning-filtered price series — Boeing (BA).

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

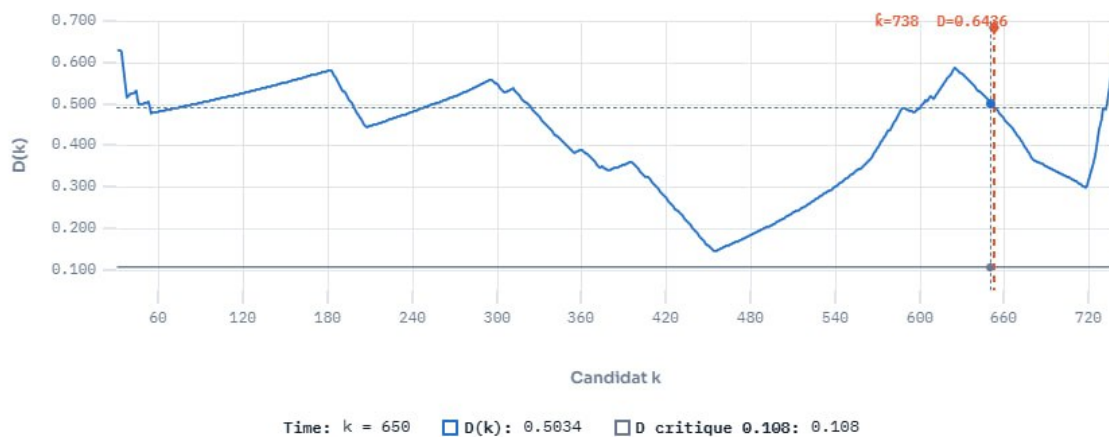
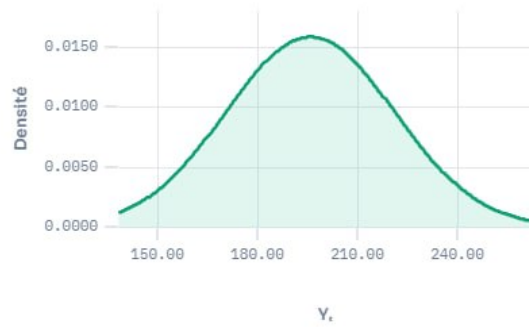


Figure 3.20: KS scan — Boeing (BA). Estimated break: $\hat{k} = 738$ (2025-12-15).

3 Ajustement Skew-Normal par Régime

$$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z), \theta \text{ partagé.}$$

Régime 1



Time: -- ■ SN $\mu=195.8357$ $\sigma=25.2167$ $\theta=0.000$: --

Régime 2



Time: -- ■ SN $\mu=229.8417$ $\sigma=15.1891$ $\theta=0.000$: --

Figure 3.21: Regime histograms with fitted SN densities — Boeing (BA).

3.3.4 BAE Systems (BAESY)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4} X_{t-1} + \frac{1}{2} X_t + \frac{1}{4} X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.



Figure 3.22: Raw and Hanning-filtered price series — BAE Systems (BAESY).

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

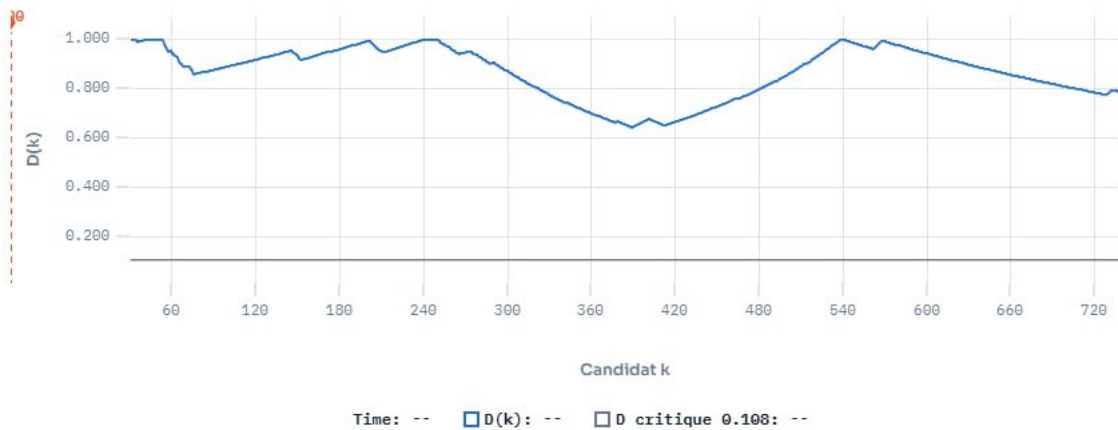
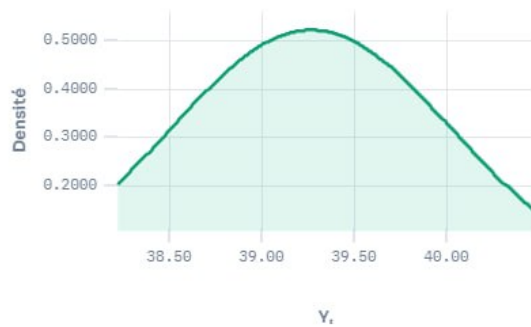


Figure 3.23: KS scan — BAE Systems (BAESY). Estimated break: $\hat{k} = 30$ (2023-02-17).

3 Ajustement Skew-Normal par Régime

$$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z), \theta \text{ partagé.}$$

Régime 1



Régime 2

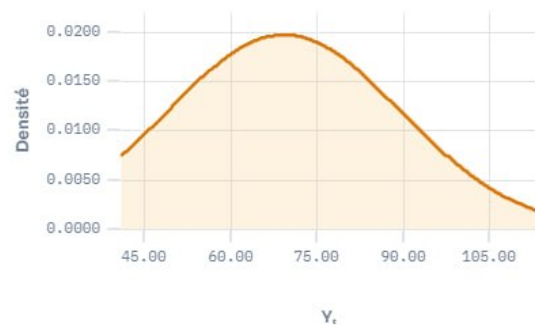


Figure 3.24: Regime histograms with fitted SN densities — BAE Systems (BAESY).

3.3.5 Bombardier Inc. (BDRAF)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.



Figure 3.25: Raw and Hanning-filtered price series — Bombardier (BDRAF).

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

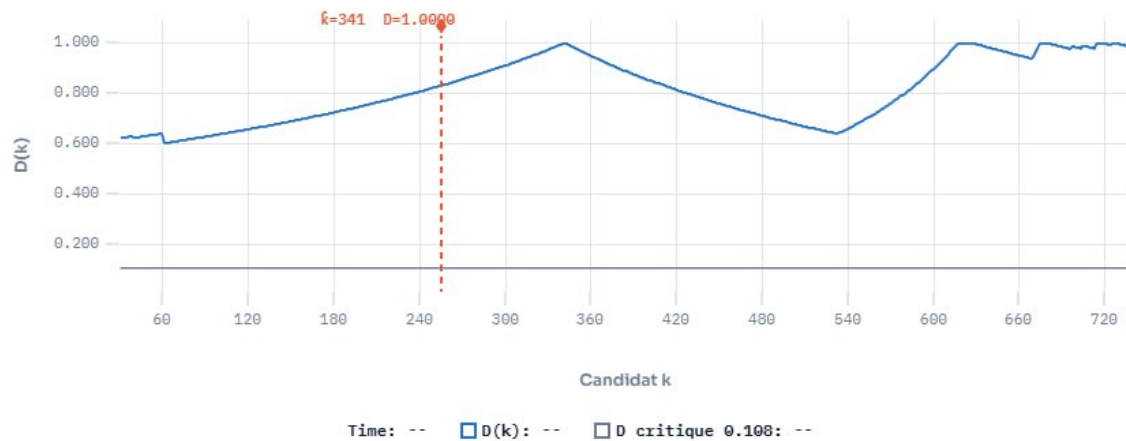
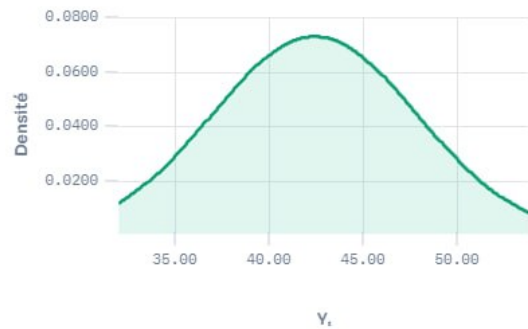


Figure 3.26: KS scan — Bombardier (BDRAF). Estimated break: $\hat{k} = 341$ (2024-05-15).

3 Ajustement Skew-Normal par Régime

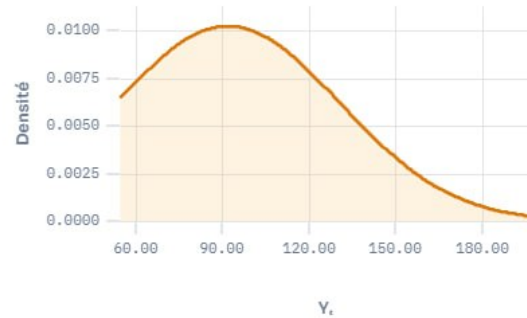
$$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z), \theta \text{ partagé.}$$

Régime 1



Time: -- ■ SN $\mu=42.4406$ $\sigma=5.4676$ $\theta=0.000$: --

Régime 2



Time: -- ■ SN $\mu=91.8179$ $\sigma=38.8712$ $\theta=0.000$: --

Figure 3.27: Regime histograms with fitted SN densities — Bombardier (BDRAF).

2 Y_t (Hanning) coloré par régime

Vert = Régime 1 ($t < \hat{k}$). Ambre = Régime 2 ($t \geq \hat{k}$).

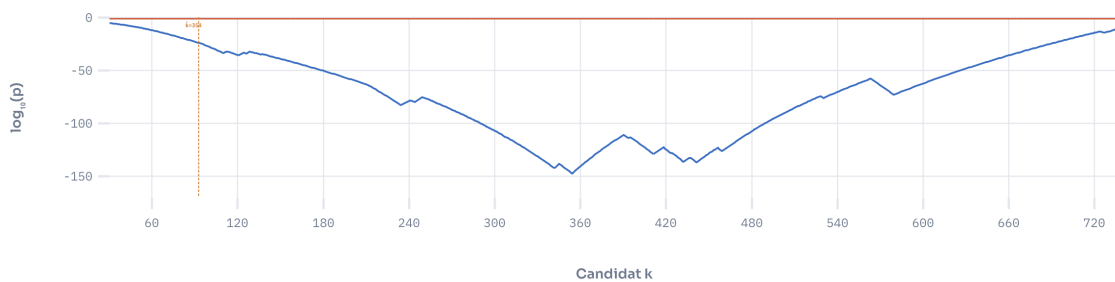


Time: -- ■ Régime 1 n=341: -- ■ Régime 2 n=427: --

Figure 3.28: Scan KS exhaustif — D(k)

5 Scan p-valeur — $\log_{10}(p(k))$

Axe en $\log_{10}$ pour la lisibilité. Ligne rouge = $\alpha = 5\%$. Ligne ambre = $\hat{k}$.



Time: -- ■ $\log_{10}(p\text{-valeur})$: -- ■ $\alpha = 0.05$ (-1.30): --

Figure 3.29: p-valeur

6 Comparaison ECDF

Segment rouge = écart maximal D (statistique KS).

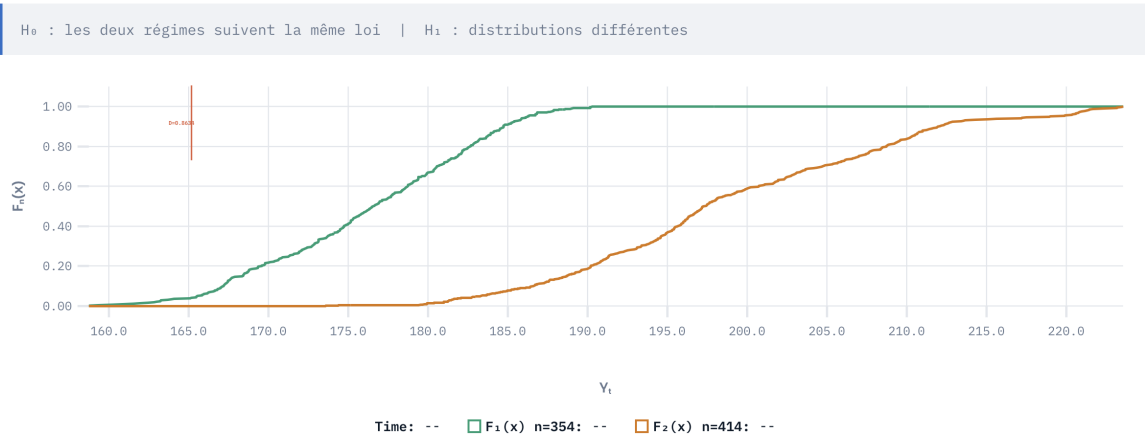


Figure 3.30: Comparaison ECDF

7 Moyenne Mobile (h = 15)

$$M_h(t) = \frac{1}{2h+1} \sum_{j=-h}^h X_{t+j}$$

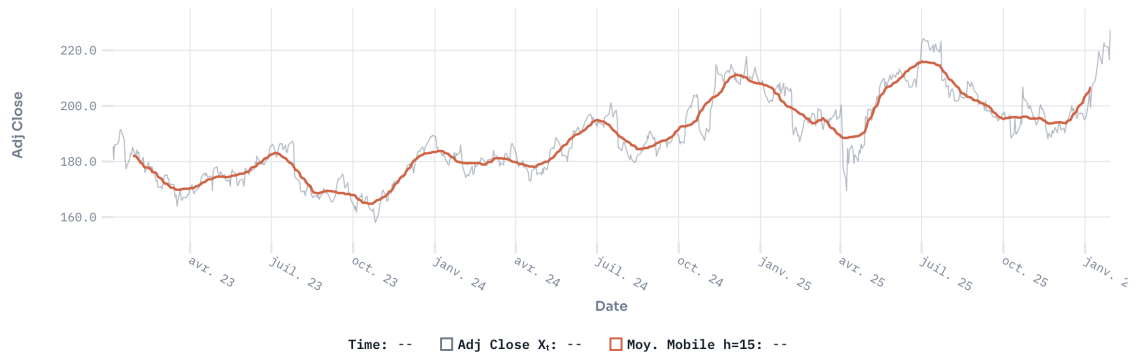


Figure 3.31: Moyenne Mobile

8 Statistiques descriptives — Adj Close

Table des statistiques descriptives pour : Honeywell International Inc.

STATISTIQUE	VALEUR
Nombre d'observations	770
Minimum	158.0669
Maximum	227.2400
Moyenne empirique	188.5857
Variance empirique	208.3680
Skewness (γ_1)	0.3190
Kurtosis (γ_2)	-0.6445

Figure 3.32: Statistiques descriptives — Adj Close

9 Paramètres MLE — Résumé

$$\hat{\theta}, \hat{\mu}_1, \hat{\sigma}_1, \hat{\mu}_2, \hat{\sigma}_2 = \arg \max [\ell_1(\mu_1, \sigma_1, \theta) + \ell_2(\mu_2, \sigma_2, \theta)]$$

RÉGIME	N	M	Σ	θ PARTAGÉ	-LOG L
Régime 1	354	176.286289	6.680706	0.000000	2722.51
Régime 2	414	199.025783	10.174685		

Résultats du test KS

PARAMÈTRE	VALEUR
Indice de rupture $\hat{k}$	354
Date de rupture	2024-06-04
Statistique D	0.863424
p-valeur	7.1869e-148
Décision ($\alpha = 5\%$)	Reject H_0
n_1 (avant rupture)	354
n_2 (après rupture)	414

Figure 3.33: Paramètres MLE — Résumé

3.3.6 General Dynamics (GD)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.

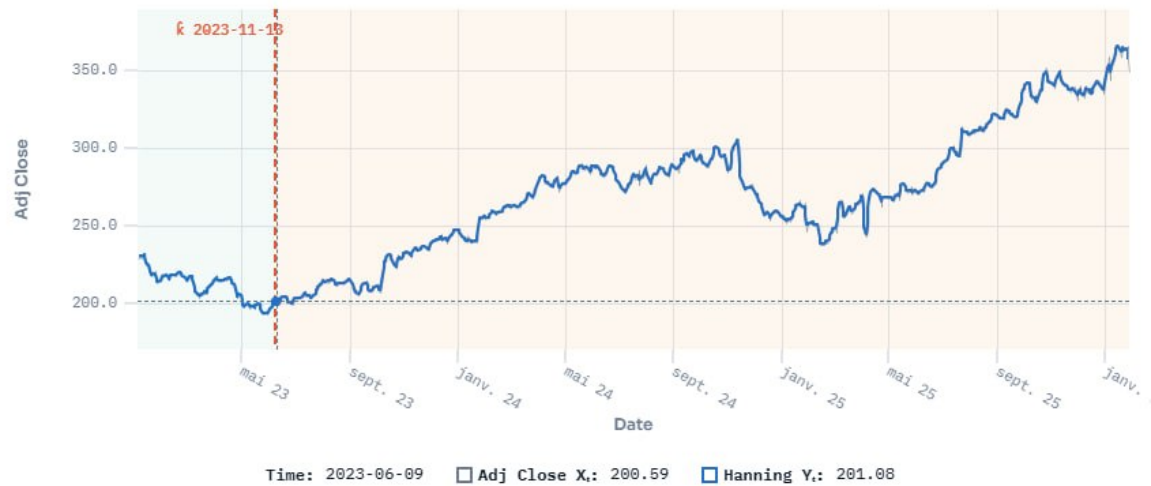


Figure 3.34: Raw and Hanning-filtered price series — General Dynamics (GD).

4 Scan KS exhaustif — D(k)

D(k) pour chaque candidat k. Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

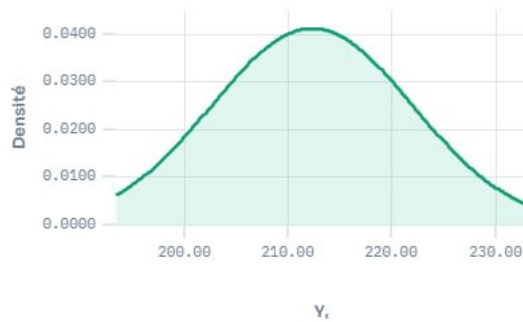


Figure 3.35: KS scan — General Dynamics (GD). Estimated break: $\hat{k} = 215$ (2023-11-13).

3 Ajustement Skew-Normal par Régime

$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z)$, θ partagé.

Régime 1



Régime 2



Figure 3.36: Regime histograms with fitted SN densities — General Dynamics (GD).

3.3.7 Honeywell International (HON)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.

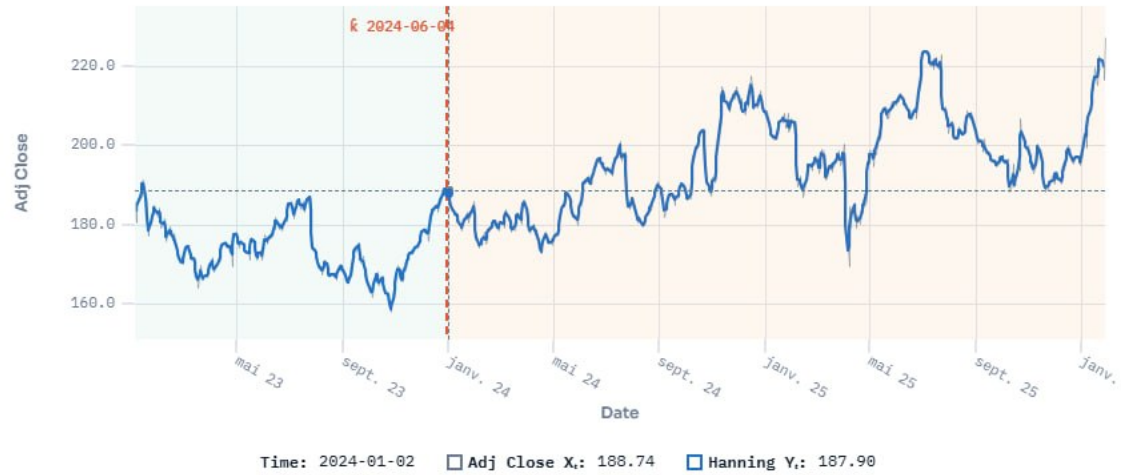


Figure 3.37: Raw and Hanning-filtered price series

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

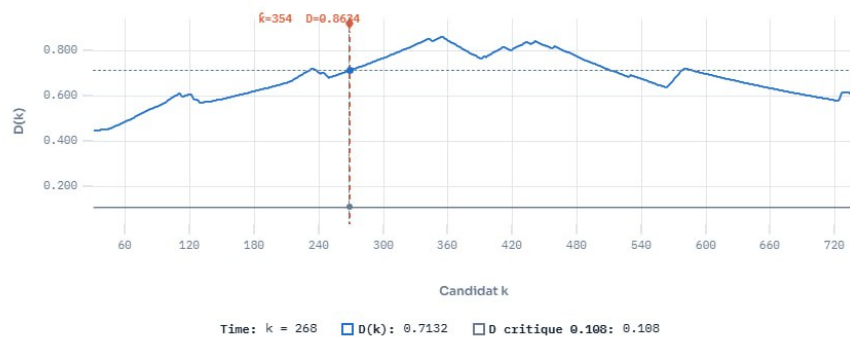
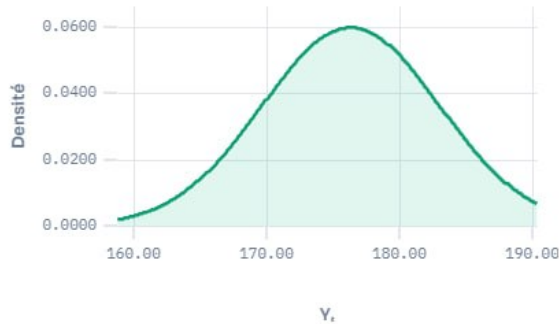


Figure 3.38: Coloré par régime

3 Ajustement Skew-Normal par Régime

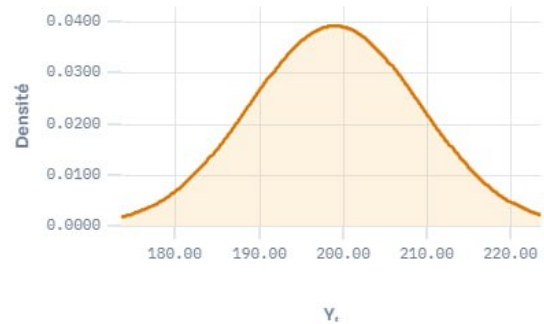
$$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z), \theta \text{ partagé.}$$

Régime 1



Time: -- ■ SN $\mu=176.2863$ $\sigma=6.6897$ $\theta=0.000$: --

Régime 2

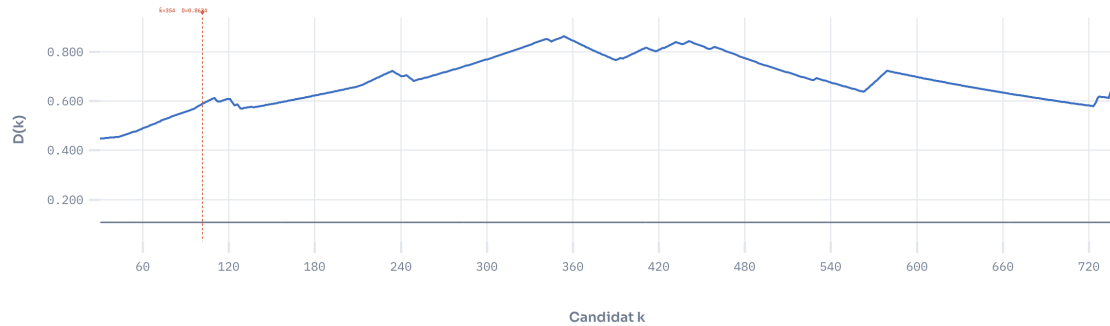


Time: -- ■ SN $\mu=199.0258$ $\sigma=10.1747$ $\theta=0.000$: --

Figure 3.39: Ajustement Skew-Normal par Régime

4 Scan KS exhaustif — D(k)

D(k) pour chaque candidat k. Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

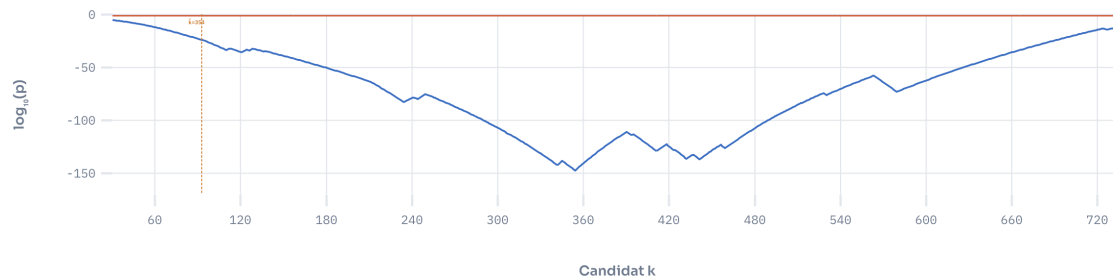


Time: -- ■ D(k): -- ■ D critique 0.108: --

Figure 3.40: Scan KS exhaustif — D(k)

5 Scan p-valeur — $\log_{10}(p(k))$

Axe en $\log_{10}$ pour la lisibilité. Ligne rouge = $\alpha = 5\%$. Ligne ambre = $\hat{k}$.



Time: -- ■ $\log_{10}(p\text{-valeur})$: -- ■ $\alpha = 0.05$ (-1.30): --

Figure 3.41: p-valeur

6 Comparaison ECDF

Segment rouge = écart maximal D (statistique KS).

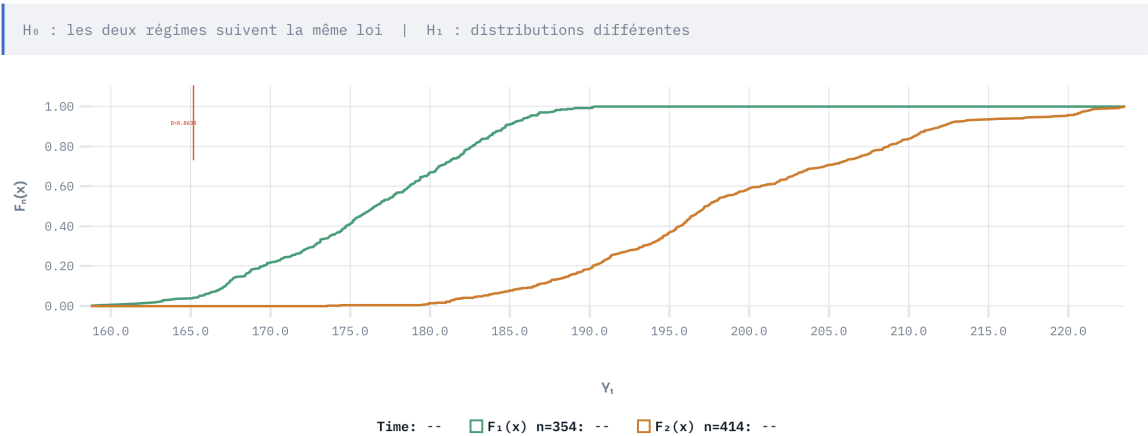


Figure 3.42: Comparaison ECDF

7 Moyenne Mobile (h = 15)

$$M_h(t) = \frac{1}{2h+1} \sum_{j=-h}^h X_{t+j}$$

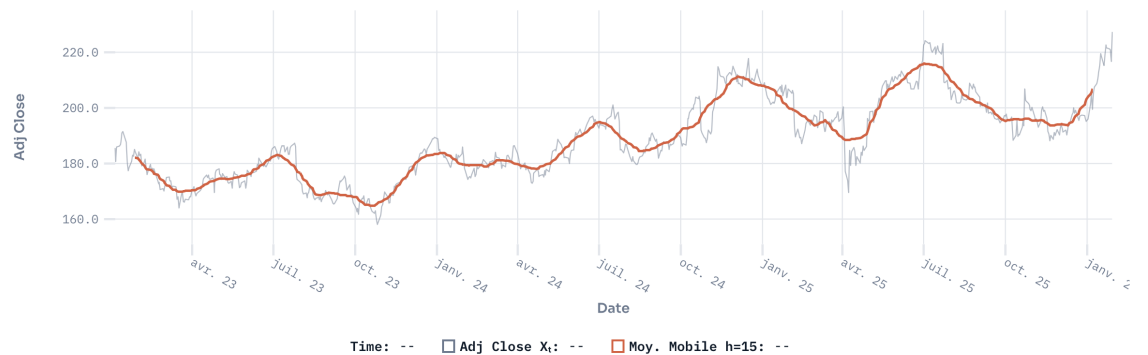


Figure 3.43: Moyenne Mobile

8 Statistiques descriptives — Adj Close

Table des statistiques descriptives pour : Honeywell International Inc.

STATISTIQUE	VALEUR
Nombre d'observations	770
Minimum	158.0669
Maximum	227.2400
Moyenne empirique	188.5857
Variance empirique	208.3680
Skewness (γ_1)	0.3190
Kurtosis (γ_2)	-0.6445

Figure 3.44: Statistiques descriptives — Adj Close

9 Paramètres MLE — Résumé

$$\hat{\theta}, \hat{\mu}_1, \hat{\sigma}_1, \hat{\mu}_2, \hat{\sigma}_2 = \arg \max [\ell_1(\mu_1, \sigma_1, \theta) + \ell_2(\mu_2, \sigma_2, \theta)]$$

RÉGIME	N	M	Σ	θ PARTAGÉ	-LOG L
Régime 1	354	176.286289	6.680706	0.000000	2722.51
Régime 2	414	199.025783	10.174685		

Résultats du test KS

PARAMÈTRE	VALEUR
Indice de rupture $\hat{k}$	354
Date de rupture	2024-06-04
Statistique D	0.863424
p-valeur	7.1869e-148
Décision ($\alpha = 5\%$)	Reject H_0
n_1 (avant rupture)	354
n_2 (après rupture)	414

Figure 3.45: Paramètres MLE — Résumé

3.3.8 Lockheed Martin (LMT)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.



Figure 3.46: Raw and Hanning-filtered price series

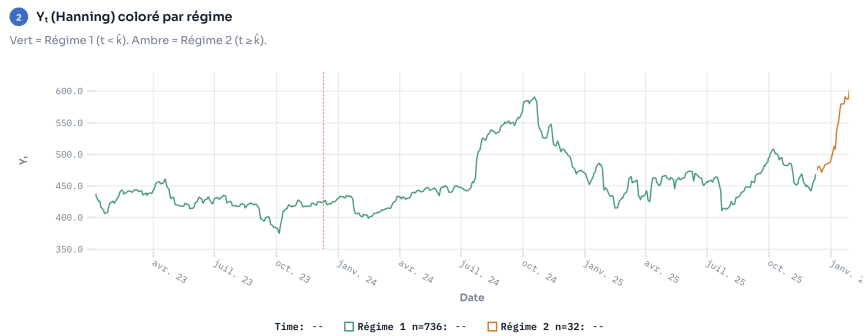
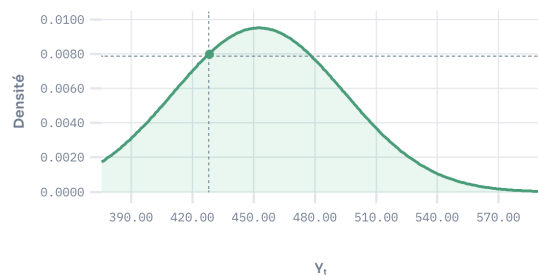


Figure 3.47: Coloré par régime

3 Ajustement Skew-Normal par Régime

$$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z), \theta \text{ partagé.}$$

Régime 1



Régime 2

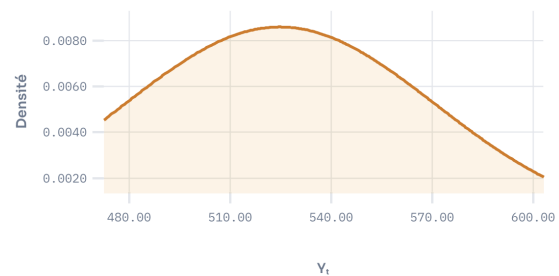


Figure 3.48: Ajustement Skew-Normal par Régime

4 Scan KS exhaustif — D(k)

D(k) pour chaque candidat k. Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 (α = 5 %).

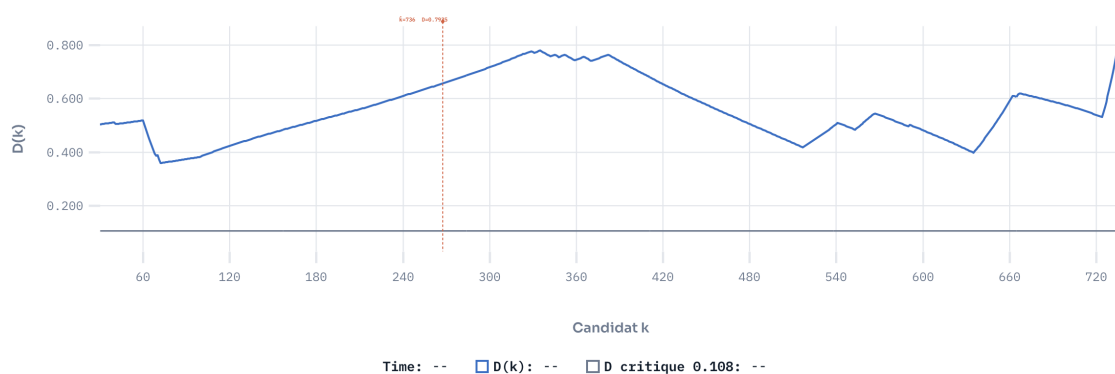


Figure 3.49: Scan KS exhaustif — D(k)

5 Scan p-valeur — $\log_{10}(p(k))$

Axe en $\log_{10}$ pour la lisibilité. Ligne rouge = $\alpha = 5\%$. Ligne ambre = $\hat{k}$.

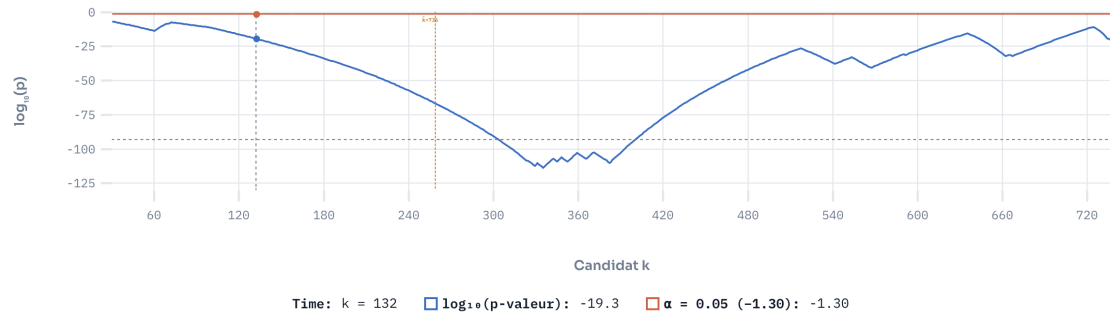


Figure 3.50: p-valeur

6 Comparaison ECDF

Segment rouge = écart maximal D (statistique KS).

H_0 : les deux régimes suivent la même loi | H_1 : distributions différentes

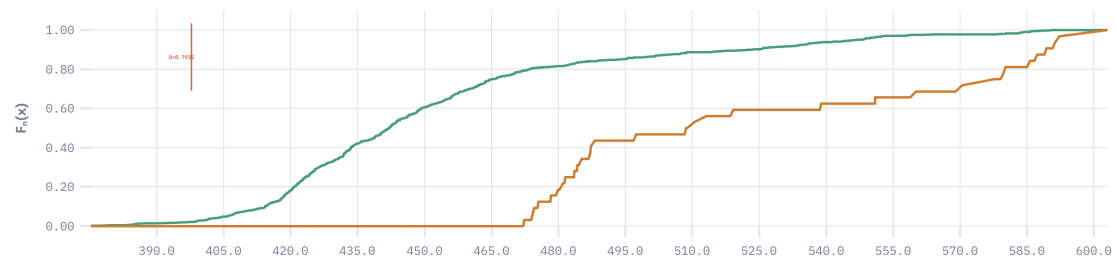


Figure 3.51: Comparaison ECDF

7 Moyenne Mobile (h = 15)

$$M_h(t) = \frac{1}{2h+1} \sum_{j=-h}^h X_{t+j}$$

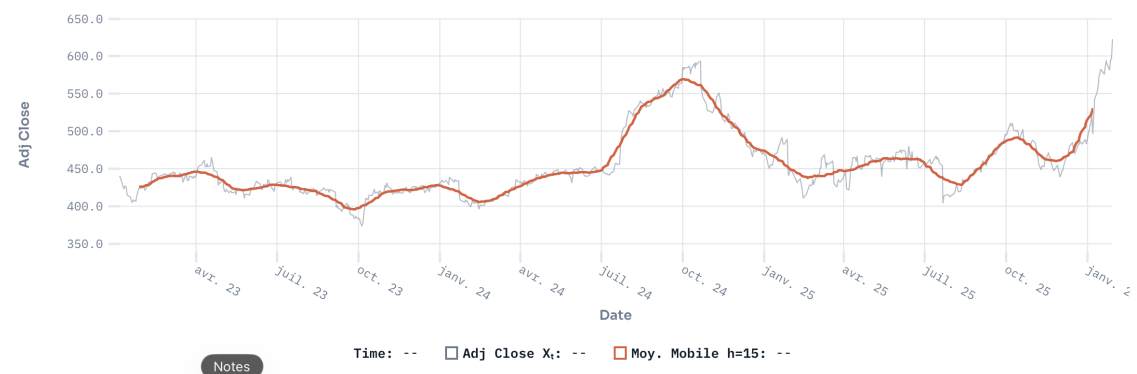


Figure 3.52: Moyenne Mobile

8 Statistiques descriptives — Adj Close

Table des statistiques descriptives pour : Lockheed Martin Corporation

STATISTIQUE	VALEUR
Nombre d'observations	770
Minimum	373.4631
Maximum	622.5100
Moyenne empirique	455.5700
Variance empirique	2020.6153
Skewness (γ_1)	1.2655
Kurtosis (γ_2)	1.1977

Figure 3.53: Statistiques descriptives — Adj Close

9 Paramètres MLE — Résumé

$$\hat{\theta}, \hat{\mu}_1, \hat{\sigma}_1, \hat{\mu}_2, \hat{\sigma}_2 = \arg \max [\ell_1(\mu_1, \sigma_1, \theta) + \ell_2(\mu_2, \sigma_2, \theta)]$$

RÉGIME	N	M	Σ	θ PARTAGÉ	-LOG L
Régime 1	736	452.371406	41.853773	0.000000	3960.84
Régime 2	32	524.626719	46.316047		

Résultats du test KS

PARAMÈTRE	VALEUR
Indice de rupture $\hat{k}$	736
Date de rupture	2025-12-11
Statistique D	0.793478
p-valeur	7.3876e-21
Décision ($\alpha = 5\%$)	Reject H_0
n_1 (avant rupture)	736
n_2 (après rupture)	32

Figure 3.54: Paramètres MLE — Résumé

3.3.9 Northrop Grumman (NOC)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.



Figure 3.55: Raw and Hanning-filtered price series

2 Y_t (Hanning) coloré par régime

Vert = Régime 1 ($t < \hat{k}$). Orange = Régime 2 ($t \geq \hat{k}$).

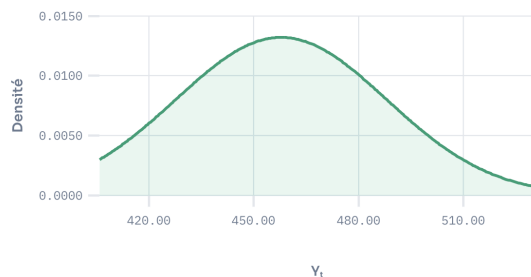


Figure 3.56: Coloré par régime

3 Ajustement Skew-Normal par Régime

$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z)$, θ partagé.

Régime 1



Régime 2

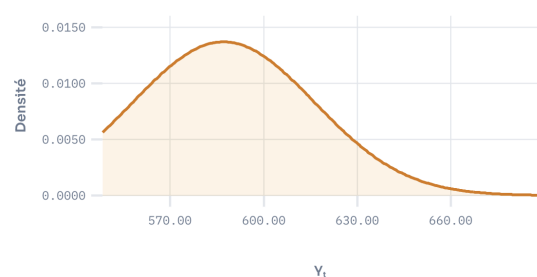


Figure 3.57: Ajustement Skew-Normal par Régime

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

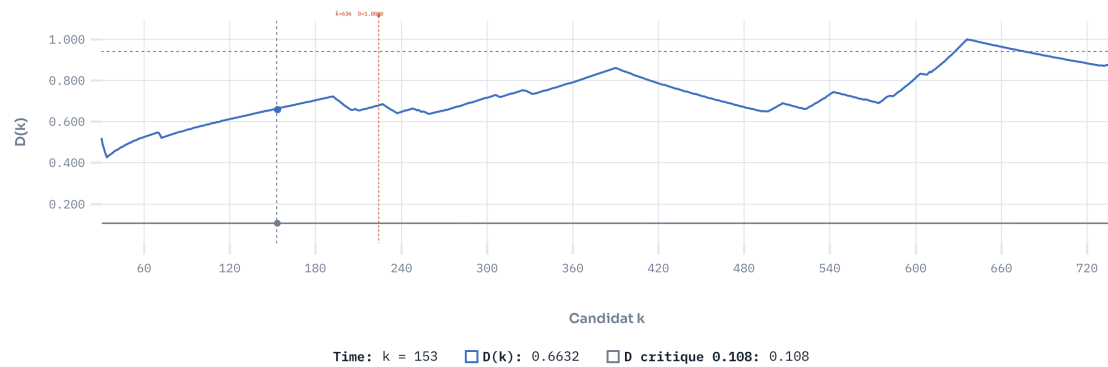


Figure 3.58: Scan KS exhaustif — $D(k)$

5 Scan p-valeur — $\log_{10}(p(k))$

Axe en $\log_{10}$ pour la lisibilité. Ligne rouge = $\alpha = 5\%$. Ligne ambre = $\hat{k}$.

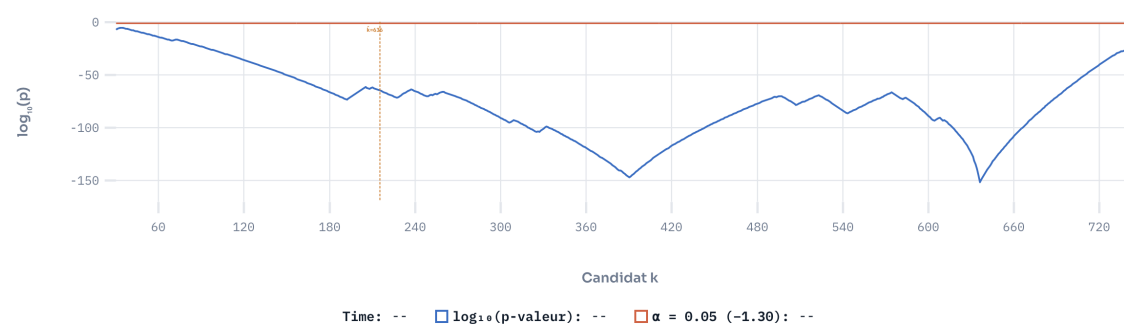


Figure 3.59: p-valeur

6 Comparaison ECDF

Segment rouge = écart maximal D (statistique KS).

H_0 : les deux régimes suivent la même loi | H_1 : distributions différentes

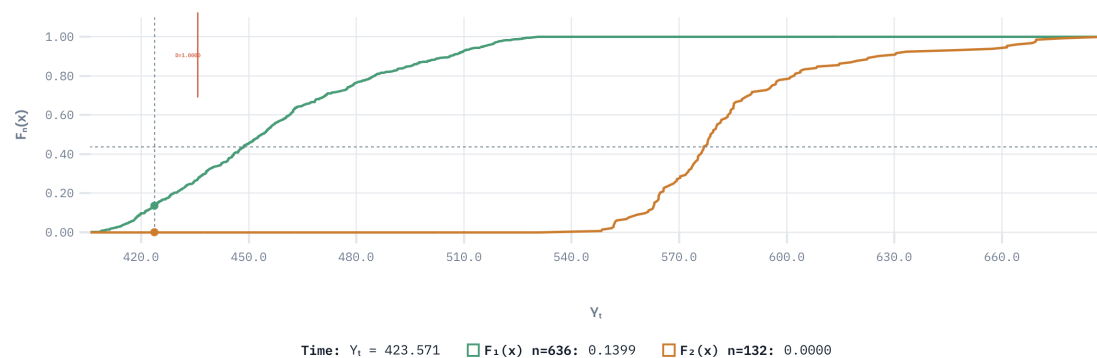


Figure 3.60: Comparaison ECDF

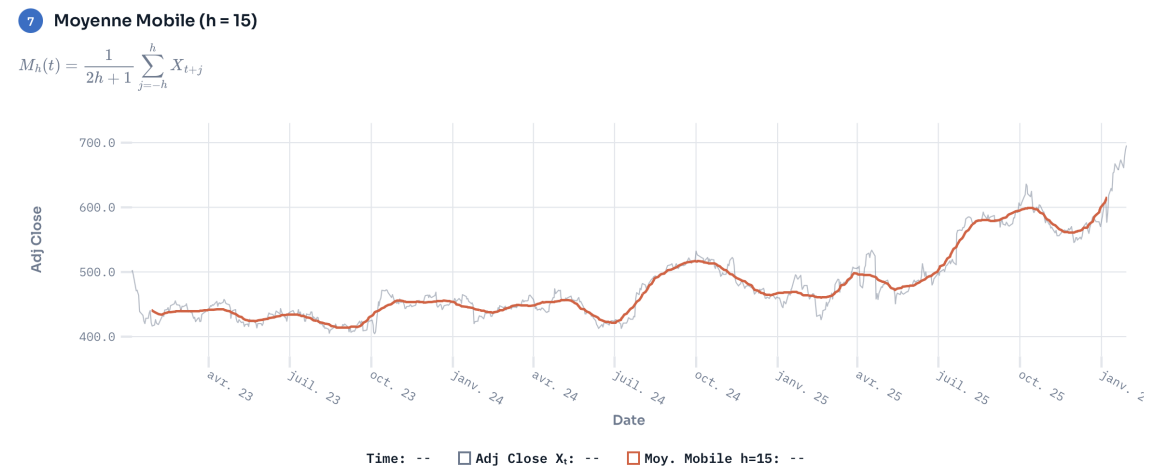


Figure 3.61: Moyenne Mobile

8 Statistiques descriptives — Adj Close

Table des statistiques descriptives pour : Northrop Grumman Corporation

STATISTIQUE	VALEUR
Nombre d'observations	770
Minimum	404.4595
Maximum	695.3500
Moyenne empirique	480.2517
Variance empirique	3349.7100
Skewness (γ_1)	1.1673
Kurtosis (γ_2)	0.7401

Figure 3.62: Statistiques descriptives — Adj Close

9 Paramètres MLE — Résumé

$\hat{\theta}, \hat{\mu}_1, \hat{\sigma}_1, \hat{\mu}_2, \hat{\sigma}_2 = \arg \max [\ell_1(\mu_1, \sigma_1, \theta) + \ell_2(\mu_2, \sigma_2, \theta)]$

RÉGIME	N	M	Σ	θ PARTAGÉ	-LOG L
Régime 1	636	457.725075	30.170459	0.000000	3701.46
Régime 2	132	587.012579	29.103346		

Résultats du test KS

PARAMÈTRE	VALEUR
Indice de rupture $\hat{k}$	636
Date de rupture	2025-07-22
Statistique D	1.000000
p-valeur	4.7512e-152
Décision ($\alpha = 5\%$)	Reject H_0
n_1 (avant rupture)	636
n_2 (après rupture)	132

Figure 3.63: Paramètres MLE — Résumé

10 Interprétation statistique

Le test KS à deux échantillons **rejette H_0** ($p = 4.751e-152 < 0.05$), confirmant une rupture structurelle au point **$\hat{k} = 636$ (2025-07-22)**.

Régime 1 — n = 636 $\mu_1 = 457.725075$ $\sigma_1 = 30.170459$	Régime 2 — n = 132 $\mu_2 = 587.012579$ $\sigma_2 = 29.103346$
--	--

θ partagé = 0.0000 → distribution asymétrique à **quasi-symétrique**.
Le rendement moyen a **augmenté** (457.7251 → 587.0126) — amélioration.
La volatilité a **diminué** (30.1705 → 29.1033) en Régime 2.

Figure 3.64: Paramètres MLE — Résumé

3.3.10 RTX Corporation (RTX)

1 Prix brut vs Filtre de Hanning

Gris = Adj Close brut (X_t). Bleu = série lissée Hanning $Y_t = \frac{1}{4}X_{t-1} + \frac{1}{2}X_t + \frac{1}{4}X_{t+1}$. Ligne rouge pointillée = rupture $\hat{k}$.

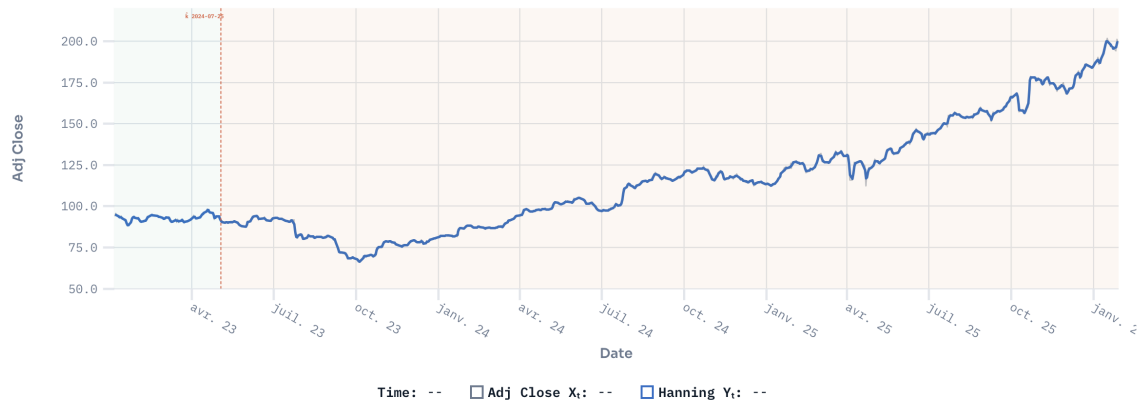


Figure 3.65: Raw and Hanning-filtered price series

2 Y_t (Hanning) coloré par régime

Vert = Régime 1 ($t < \hat{k}$). Ambre = Régime 2 ($t \geq \hat{k}$).

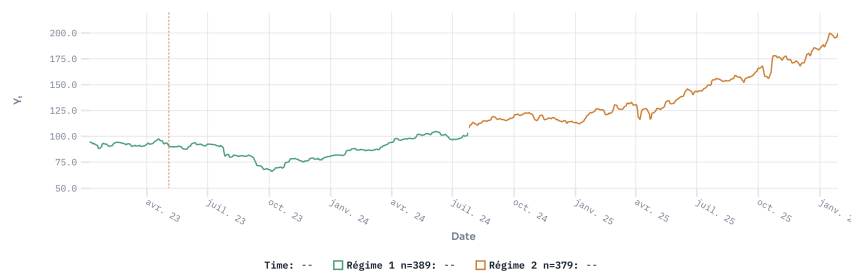
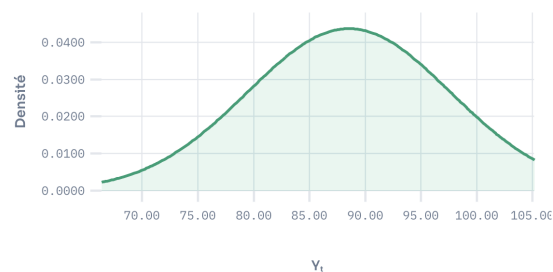


Figure 3.66: Coloré par régime

3 Ajustement Skew-Normal par Régime

$f(x; \mu, \sigma, \theta) = \frac{2}{\sigma} \varphi(z) \Phi(\theta z)$, θ partagé.

Régime 1



Régime 2

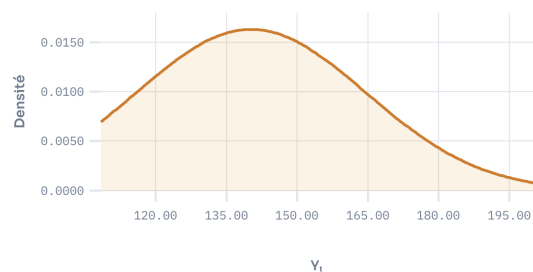


Figure 3.67: Ajustement Skew-Normal par Régime

4 Scan KS exhaustif — $D(k)$

$D(k)$ pour chaque candidat k . Losange rouge = $\hat{k}$. Ligne grise = D critique 0.108 ($\alpha = 5\%$).

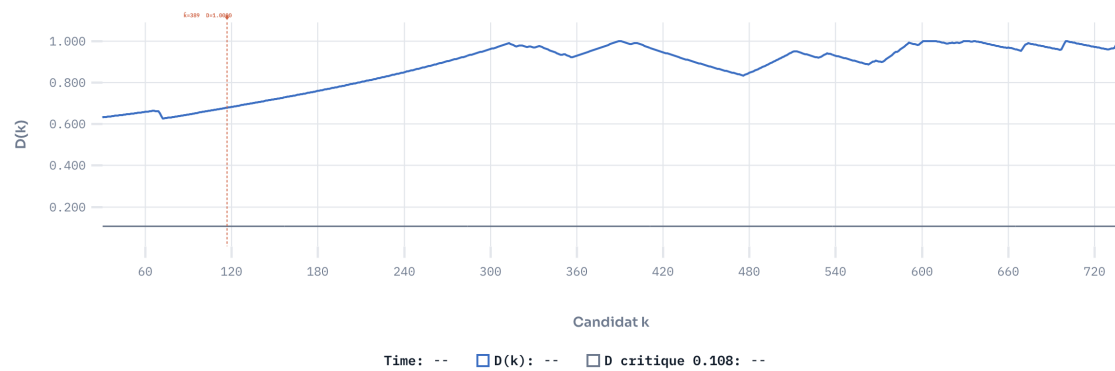


Figure 3.68: Scan KS exhaustif — $D(k)$

5 Scan p-valeur — $\log_{10}(p(k))$

Axe en $\log_{10}$ pour la lisibilité. Ligne rouge = $\alpha = 5\%$. Ligne ambre = $\hat{k}$.

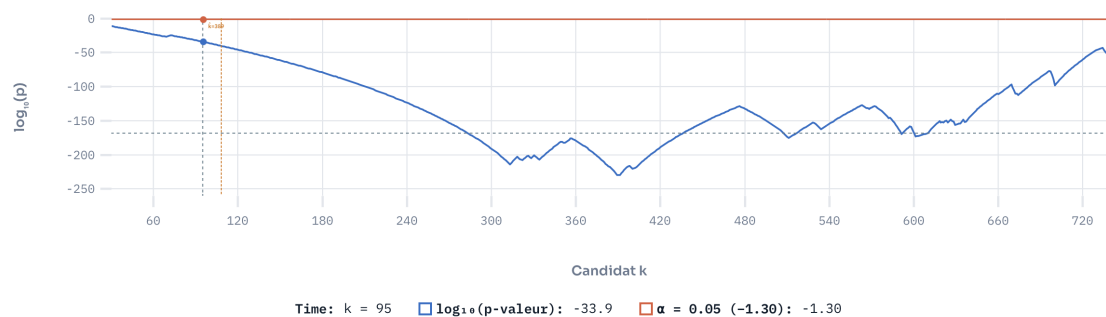


Figure 3.69: p-valeur

6 Comparaison ECDF

Segment rouge = écart maximal D (statistique KS).

H_0 : les deux régimes suivent la même loi | H_1 : distributions différentes

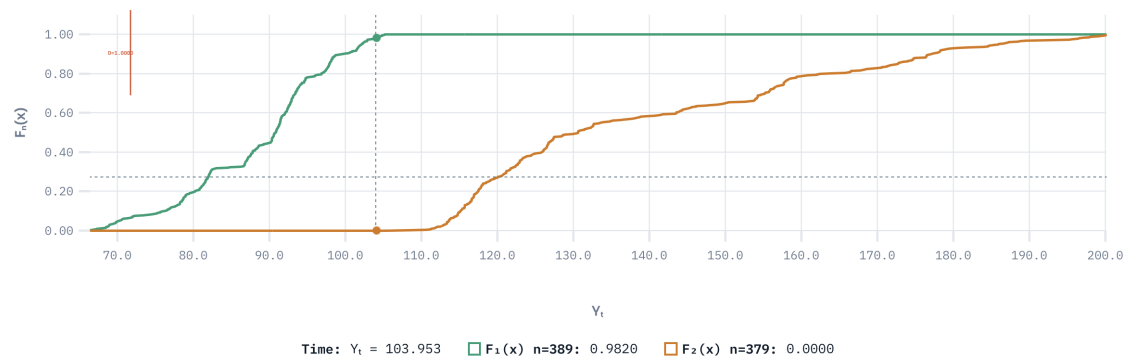


Figure 3.70: Comparaison ECDF

7 Moyenne Mobile (h = 15)

$$M_h(t) = \frac{1}{2h+1} \sum_{j=-h}^h X_{t+j}$$

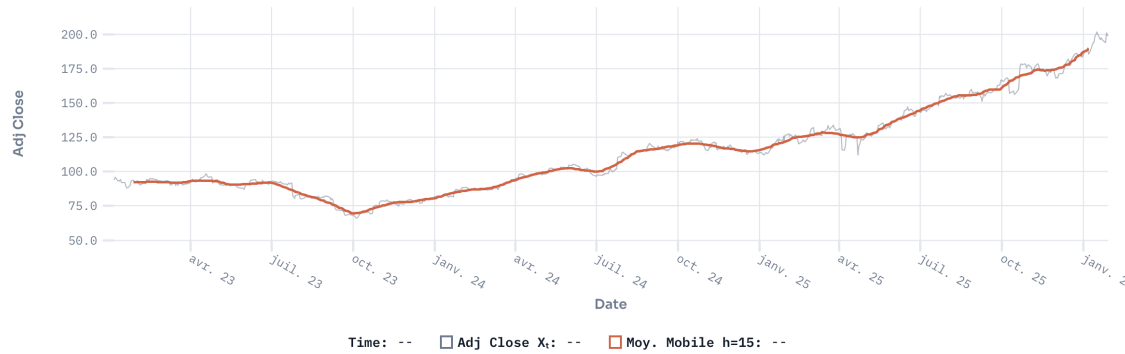


Figure 3.71: Moyenne Mobile

8 Statistiques descriptives — Adj Close

Table des statistiques descriptives pour : RTX Corporation

STATISTIQUE	VALEUR
Nombre d'observations	770
Minimum	66.0607
Maximum	201.9200
Moyenne empirique	114.0671
Variance empirique	1011.2512
Skewness (γ ₁)	0.8041
Kurtosis (γ ₂)	-0.2141

Figure 3.72: Statistiques descriptives — Adj Close

9 Paramètres MLE — Résumé

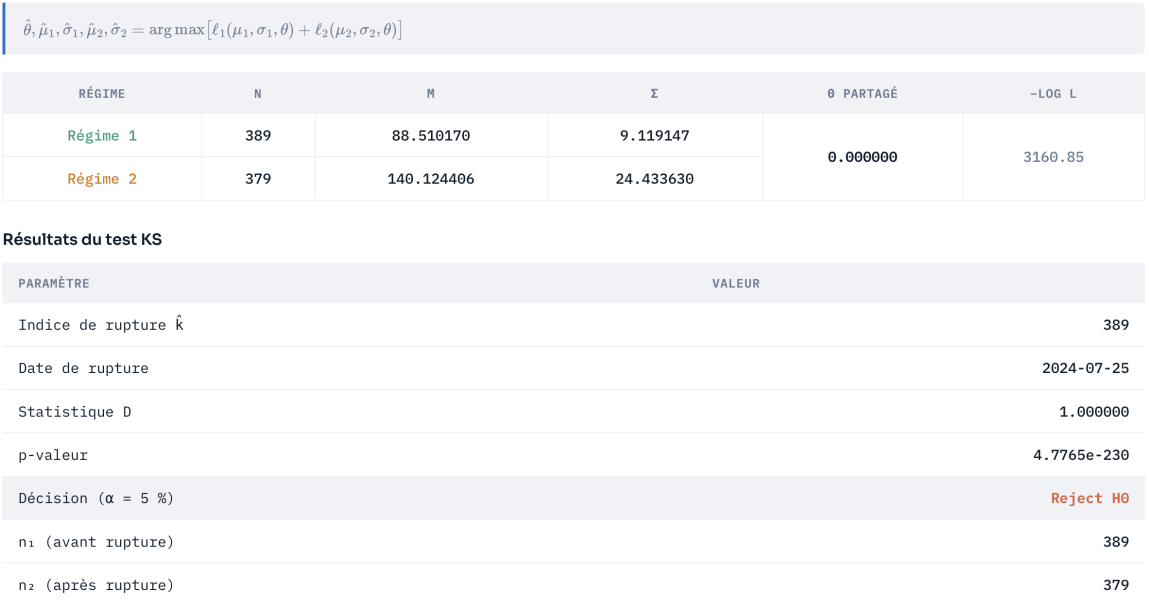


Figure 3.73: Paramètres MLE — Résumé

10 Interprétation statistique

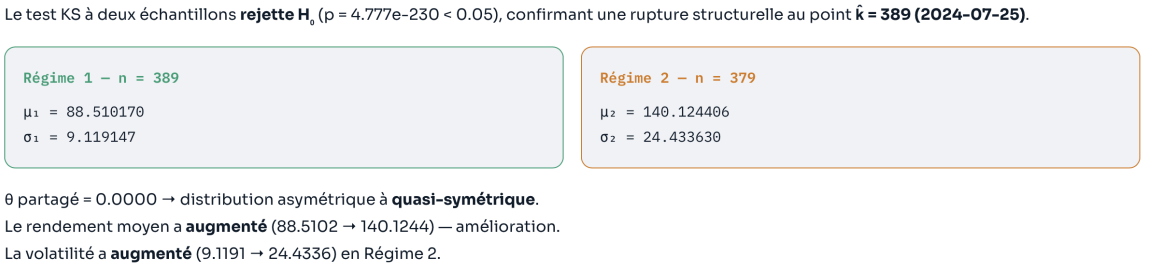


Figure 3.74: Paramètres MLE — Résumé

3.4 Analysis and Interpretation

3.4.1 Universal Rejection of H_0

The KS test **rejects the null hypothesis** for all ten companies at the 5% level — in most cases with a p-value effectively equal to zero. This provides overwhelming evidence that every aerospace stock in our sample underwent at least one distributional regime change between January 2023 and the end of the observation window.

3.4.2 Company-by-Company Discussion

Airbus (AIR.PA). The estimated break falls at index $\hat{k} = 628$ (24 June 2025), near the end of the sample. The location parameter rises from $\hat{\mu}_1 = 137.12$ to $\hat{\mu}_2 = 194.33$ (+41.7%), while the scale decreases from $\hat{\sigma}_1 = 16.53$ to $\hat{\sigma}_2 = 12.58$, indicating a higher and less dispersed price regime in the final months.

Dassault Aviation (AM.PA). Break at 21 January 2025 ($\hat{k} = 521$). The mean price increases by nearly 59 % across regimes ($177 \rightarrow 282$), with substantially higher volatility in the second regime ($\hat{\sigma}_2 = 27.71$ vs. $\hat{\sigma}_1 = 13.32$).

Boeing (BA). Break detected very late in the sample ($\hat{k} = 738$, 15 December 2025), with only 30 post-break observations. This suggests the major shift was still unfolding at the end of the window; Boeing’s ongoing production and regulatory challenges are a plausible driver.

BAE Systems (BAESY). Unusually early break at $\hat{k} = 30$ (17 February 2023), giving only 30 pre-break observations. The scale explodes from $\hat{\sigma}_1 = 0.76$ to $\hat{\sigma}_2 = 20.26$, implying the series was nearly constant at the very start of 2023 before entering a high-volatility growth regime. This may reflect thin initial trading volume in the ADR.

Bombardier (BDRAF). Break in May 2024 ($\hat{k} = 341$). Mean roughly doubles from 42 to 92, accompanied by a large increase in dispersion. Bombardier’s restructuring and return to profitability after divesting commercial aviation could explain this positive regime shift.

General Dynamics (GD). Break in November 2023 ($\hat{k} = 215$). Mean rises from 212 to 286 (+34.7 %) with a substantial increase in variance. The timing corresponds to a period of elevated global defence spending.

Honeywell (HON). Break in June 2024 ($\hat{k} = 354$), KS statistic $D = 0.8634$. The shift is more modest (+13 % in mean, moderate variance increase), consistent with Honeywell’s diversified industrial profile buffering large shocks.

Lockheed Martin (LMT). Like Boeing, the break is very late ($\hat{k} = 736$, December 2025) with few post-break observations. Mean increases from 452 to 525 (+16 %).

Northrop Grumman (NOC). Break in July 2025 ($\hat{k} = 636$). Mean jumps from 458 to 587 (+28 %), consistent with strong backlog growth in space and missile defence.

RTX Corporation (RTX). Break in July 2024 ($\hat{k} = 389$). Mean rises sharply from 89 to 140 (+58 %), and dispersion more than doubles ($9.1 \rightarrow 24.4$). RTX’s settlement of the GTF engine powder-metal disk issue in 2023 and subsequent recovery may underlie this transition.

3.4.3 Estimated Skewness

All estimated $\hat{\theta}$ values are effectively zero (or converge exactly to zero under L-BFGS-B). This implies that, after Hanning filtering, each regime is well described by a near-symmetric unimodal distribution — the Skew-Normal model subsumes the Normal

as a special case and the data do not support significant asymmetry. The use of the Skew-Normal remains justified as a robustness check: had skewness been present, the model would have captured it.

3.4.4 Segment Size Considerations

For most companies n_1 and n_2 are both large (several hundred observations each), ensuring reliable MLE estimates and a well-calibrated KS test. The exceptions are BA, LMT (very small $n_2 \approx 30$) and BAESY (very small $n_1 = 30$), where parameter estimates in the short segment should be interpreted with caution.

Chapter 4

Conclusion and Perspectives

4.1 Summary

This study applied a Hanning-filter–Skew-Normal breakpoint model to ten aerospace and defence stocks over the period starting January 2023. The main findings are:

1. **Universal structural breaks.** The KS homogeneity test rejects the null of no break for all ten companies, confirming that the aerospace sector experienced significant distributional regime changes during the observation window.
2. **Upward price shifts.** In every case $\hat{\mu}_2 > \hat{\mu}_1$, indicating that the post-break regime is characterised by higher prices — consistent with the sector-wide recovery and re-rating following COVID-era supply disruptions.
3. **Increased dispersion.** Most companies show $\hat{\sigma}_2 > \hat{\sigma}_1$, suggesting that the second regime is not only higher but also more uncertain.
4. **Near-zero skewness.** After smoothing with the Hanning filter, the residual skewness in each regime is negligible, validating the use of a symmetric-ish distribution for this application.

4.2 Limitations and Future Work

- **Single-break assumption.** The model allows for exactly one breakpoint. Binary segmentation or the PELT algorithm could detect multiple breaks if present.
- **Shared θ .** Allowing regime-specific skewness parameters θ_1, θ_2 would increase flexibility at the cost of identifiability.
- **Small post-break samples.** For BA and LMT the post-break window contains only ~ 30 observations. Extending the data window to mid-2026 would provide more reliable estimates.
- **Multivariate extensions.** A joint break detection across all ten stocks could identify sector-wide shock dates and distinguish idiosyncratic from systemic

breaks.

- **Returns vs. prices.** Analysing log-returns rather than filtered prices would yield stationary series and facilitate comparison with AR(1) and GARCH-type models.

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Appendix A

Core Python Implementation

A.1 Data Loading and Hanning Filter

```
1 START_DATE = "2023-01-04"
2 DATA_DIR  = "./data/Secteur_Aerospace"
3
4 def load_series(ticker):
5     import os
6     import pandas as pd
7
8     path = os.path.join(DATA_DIR, f"{ticker}.csv")
9     df = pd.read_csv(path)[["Date", "Adj Close"]].copy()
10    df.columns = ["Date", "X_t"]
11    df["Date"] = pd.to_datetime(df["Date"])
12    df = (df[df["Date"] >= pd.Timestamp(START_DATE)]
13          .sort_values("Date")
14          .dropna()
15          .reset_index(drop=True))
16    return df
17
18
19 def hanning_filter(df):
20     df = df.copy()
21     df["Y_t"] = ( 0.25 * df["X_t"].shift(1)
22                  + 0.50 * df["X_t"]
23                  + 0.25 * df["X_t"].shift(-1))
24    return df.dropna().reset_index(drop=True)
```

Listing A.1: Load adjusted closing prices and apply Hanning filter.

A.2 KS Scan for Breakpoint Estimation

```
1 def ks_scan(series, margin=30):
2     import numpy as np
3     from scipy.stats import ks_2samp
4
```

```

5     n        = len(series)
6     results = [
7         (k, *ks_2samp(series[:k], series[k:]))
8         for k in range(margin, n - margin + 1)
9     ]
10    k_values, stats, pvals = zip(*results)
11    best = int(np.argmax(stats))
12
13    return { "k_values" : list(k_values)
14            , "ks_stats" : list(stats)
15            , "p_values" : list(pvals)
16            , "k_hat"    : k_values[best]
17            , "D"        : float(stats[best])
18            , "p_value"  : float(pvals[best])
19        }

```

Listing A.2: Scan all candidate breakpoints and return the maximiser of the KS statistic.

A.3 Skew-Normal PDF and Joint MLE

```

1  def skew_normal_pdf(x, mu, sigma, theta):
2      from scipy.stats import norm
3
4      z = (x - mu) / sigma
5      return (2.0 / sigma) * norm.pdf(z) * norm.cdf(theta * z)
6
7
8  def fit_skew_normal(y1, y2):
9      import numpy as np
10     from scipy.optimize import minimize
11
12     def _neg_loglik(params):
13         mu1, log_s1, mu2, log_s2, theta = params
14         s1, s2 = np.exp(log_s1), np.exp(log_s2)
15         eps = 1e-12
16         ll = ( np.sum(np.log(np.clip(skew_normal_pdf(y1, mu1, s1,
17             theta), eps, None)))
18             + np.sum(np.log(np.clip(skew_normal_pdf(y2, mu2, s2,
19             theta), eps, None)))
20
21         return -ll
22
23     s1_init = max(float(np.std(y1, ddof=0)), 1e-3)
24     s2_init = max(float(np.std(y2, ddof=0)), 1e-3)
25     x0 = np.array([np.mean(y1), np.log(s1_init),
26                   np.mean(y2), np.log(s2_init), 0.0])
27     res = minimize(_neg_loglik, x0=x0, method="L-BFGS-B")
28
29     mu1, log_s1, mu2, log_s2, theta = res.x
30     return { "mu1" : float(mu1)

```

```

28         , "sigma1" : float(np.exp(log_s1))
29         , "mu2"     : float(mu2)
30         , "sigma2" : float(np.exp(log_s2))
31         , "theta"  : float(theta)
32     }

```

Listing A.3: Skew-Normal density and joint maximum likelihood estimation via L-BFGS-B.

A.4 Full Analysis Pipeline

```

1  def analyse(ticker):
2      import numpy as np
3      from Dic_Aerospace import Dic_Aerospace
4
5      name      = Dic_Aerospace.get(ticker, ticker)
6      raw_df    = load_series(ticker)
7      df        = hanning_filter(raw_df)
8      Y         = df["Y_t"].to_numpy()
9
10     scan      = ks_scan(Y)
11     k_hat     = scan["k_hat"]
12     break_date = df.loc[k_hat, "Date"]
13     verdict    = "Reject H0" if scan["p_value"] < 0.05 else "Fail to
14                 reject H0"
15     mle       = fit_skew_normal(Y[:k_hat], Y[k_hat:])
16
17     return { "company" : ticker
18             , "name"   : name
19             , "k_hat"  : k_hat
20             , "bp_date" : str(break_date.date())
21             , "D"      : scan["D"]
22             , "p_value" : scan["p_value"]
23             , "reject_H0" : scan["p_value"] < 0.05
24             , "verdict" : verdict
25             , **mle
26             , "n1"     : k_hat
27             , "n2"     : len(Y) - k_hat
28         }

```

Listing A.4: Main analyse(ticker) function combining all steps.